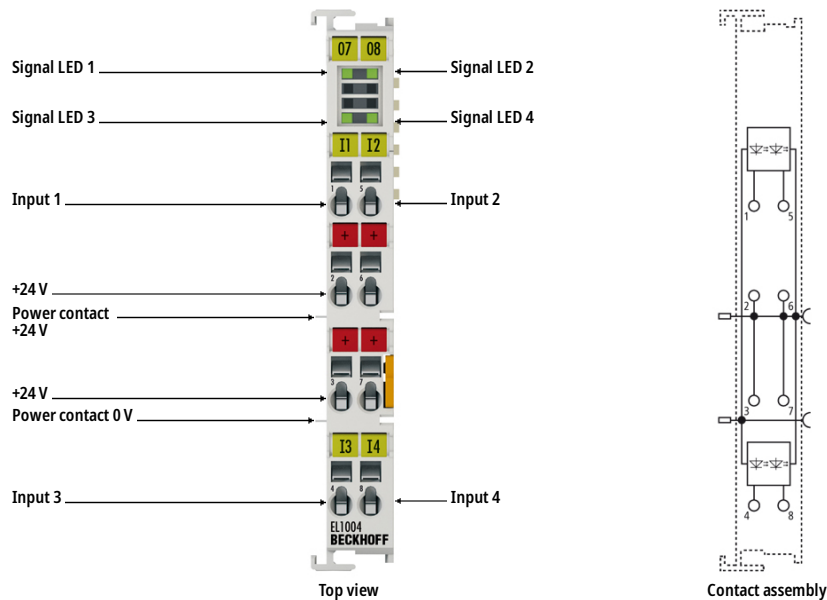




EL1004 | EtherCAT Terminal, 4-channel digital input, 24 V DC, 3 ms



i Product status: Regular delivery

The EL1004 digital input terminal acquires the binary 24 V control signals from the process level and transmits them, in an electrically isolated form, to the higher-level automation unit. The EtherCAT Terminal has four channels that indicate their signal state via light emitting diodes.

Special features:

- input specification type 1/3
- no bouncing due to mechanical switches thanks to 3 ms input filter
- direct connection of four 2-wire sensors via additional +24 V connection points

Product information

Technical Data

Technical data	EL1004, ES1004
Connection technology	2-wire
Specification	EN 61131-2, type 1/3
Number of inputs	4
Nominal voltage	24 V DC (-15 %/+20 %)
"0" signal voltage	-3...+5 V (EN 61131-2, type 3)
"1" signal voltage	11...30 V (EN 61131-2, type 3)

Input current	typ. 3 mA (EN 61131-2, type 3)
Input filter	typ. 3.0 ms
Distributed clocks	–
Current consumption power contacts	typ. 2 mA + load
Current consumption E-bus	typ. 90 mA
Electrical isolation	500 V (E-bus/field potential)
Bit width in the process image	4 inputs
Configuration	no address or configuration setting
Special features	standard input terminals for 2-wire connection
Weight	approx. 50 g
Operating/storage temperature	-25...+60 °C/-40...+85 °C
Relative humidity	95 %, no condensation
Vibration/shock resistance	conforms to EN 60068-2-6/EN 60068-2-27
EMC immunity/emission	conforms to EN 61000-6-2/EN 61000-6-4
Protect. class/installation pos.	IP 20/variable
Pluggable wiring	for all ESxxx terminals
Approvals/markings	CE, UL, ATEX, DNV GL, IECEx, cFMus
Ex marking	ATEX: II 3 G Ex nA IIC T4 Gc IECEx: Ex nA IIC T4 Gc Ex tc IIIC T135 °C Dc cFMus: Class I, Division 2, Groups A, B, C, D Class I, Zone 2, AEx ec IIC T4 Gc

Housing data	EL-12-8pin	ES-12-8pin
Design form	compact terminal housing with signal LEDs	terminal housing with pluggable wiring level
Material	polycarbonate	
Dimensions (W x H x D)	12 mm x 100 mm x 68 mm	
Installation	on 35 mm DIN rail, conforming to EN 60715 with lock	
Side by side mounting by means of	double slot and key connection	
Marking	Labeling of the BZxxx series	
Wiring	solid conductor (e), flexible conductor (f) and ferrule (a): spring actuation by screwdriver	
Connection cross-section	s*: 0.08...2.5 mm ² , st*: 0.08...2.5 mm ² , f*: 0.14...1.5 mm ²	s*: 0.08...1.5 mm ² , st*: 0.08...1.5 mm ² , f*: 0.14...1.5 mm ²
Connection cross-section AWG	s*: AWG 28...14, st*: AWG 28...14, f*: AWG 26...16	s*: AWG 28...16, st*: AWG 28...16, f*: AWG 26...16

Stripping length	8...9 mm	9...10 mm
Current load power contacts	$I_{\max}$: 10 A	

*s: solid wire; st: stranded wire; f: with ferrule

Ordering Information

Ordering information	
EL1004	EtherCAT Terminal, 4-channel digital input, 24 V DC, 3 ms
ES1004	EtherCAT Terminal, 4-channel digital input, 24 V DC, 3 ms, pluggable wiring